

# ECE4010J Probabilistic Methods in Engineering (Summer 2026)

## Assignment 6

**Date Due:** See canvas

This assignment has a total of **(100 points)**.

**Note:** Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

**Exercise 6.1 (10 pts)** Let  $X_1, \dots, X_m$  be a random sample of size  $m$  from a binomial distribution with parameters  $n$ , assumed to be known, and  $\theta$ . Show that the method of moments estimator for  $\theta$  is  $\hat{\theta} = \bar{X}/n$ .

**Exercise 6.2 (10 pts)** Find the method of moments estimator for the parameter  $\theta$  of a geometric distribution.

**Exercise 6.3 (10 pts)** Let  $X$  be normal with mean  $\mu$  and variance  $\sigma^2$ , both of which are unknown. Find the method of moments estimators for these parameters. Are the estimators obtained unbiased for their respective parameters? Explain.

**Exercise 6.4 (10 pts)** Let  $W$  be an exponential random variable with parameter  $\beta$  unknown. Find the maximum likelihood estimator for  $\beta$  based on a sample of size  $n$ . Does it differ from the method of moments estimator?

**Exercise 6.5 (10 pts)** A new material is being tested for possible use in the brake shoes of automobiles. These shoes are expected to last for at least 75,000 miles. Fifteen sets of four of these experimental shoes are subjected to accelerated life testing. The random variable  $X$ , the number of shoes in each group of four that fail early, is assumed to be binomially distributed with  $n = 4$  and  $\theta$  unknown. Find the maximum likelihood estimate for  $\theta$  based on these data:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 2 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 |   |

If an early failure rate in excess of 10% is unacceptable from a business point of view, would you have some doubts concerning the use of this new material? Explain.

**Exercise 6.6 (25 pts)** When programming from a terminal, one random variable of concern is the response time in seconds. These data are obtained for one particular installation:

|      |      |      |      |      |      |
|------|------|------|------|------|------|
| 1.48 | 1.26 | 1.52 | 1.56 | 1.48 | 1.46 |
| 1.30 | 1.28 | 1.43 | 1.43 | 1.55 | 1.57 |
| 1.51 | 1.53 | 1.68 | 1.37 | 1.47 | 1.61 |
| 1.49 | 1.43 | 1.64 | 1.51 | 1.60 | 1.65 |
| 1.60 | 1.64 | 1.51 | 1.51 | 1.53 | 1.74 |

- (i) (5pts) Construct a stem-and-leaf diagram. Does the assumption of normality appear reasonable?
- (ii) (5pts) Find the unbiased point estimate for  $\sigma^2$ .
- (iii) (5pts) Find a 95% confidence interval on  $\sigma^2$ .
- (iv) (5pts) Find a 95% confidence interval on  $\sigma$ .
- (v) (5pts) Would you be surprised to hear the director of this installation claim that the standard deviation in response time is more than 0.2 second? Explain.

**Exercise 6.7 (10 pts)** (One-sided confidence interval on  $\sigma^2$ .) Since variance is a measure of consistency, it is usually hoped that  $\sigma^2$  will be small. For this reason, it is sometimes useful to construct what is called a *one-sided* confidence interval for  $\sigma^2$ . That is, we want to find an interval of the form  $[0, L]$ , where  $L$  is a statistic with the property that  $\Pr[\sigma^2 \leq L]$ . The formula for such an interval is

$$L = (n - 1)S^2/\chi_{1-\alpha}^2$$

The point  $\chi_{1-\alpha}^2$  is the lower-tailed chi-squared point with  $\alpha$  area to its left and  $1 - \alpha$  to the right. For example, to construct a 95% one-sided confidence interval on a the chi-squared point used would be that with  $n - 1$  degrees of freedom and 0.05 area to the left. Use these data on  $X$ , the actual length of 63-mm nails, to find a 95% one-sided confidence interval on the variance in length:

|      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|
| 63.0 | 63.1 | 63.0 | 63.0 | 62.9 | 63.0 | 63.0 |
| 63.1 | 62.8 | 63.1 | 63.1 | 63.0 | 62.9 | 63.2 |

The manufacturer wants to check to be sure that the population variance of the nails being produced does not exceed 0.03. Does this sample indicate that this is the case? Explain.

**Exercise 6.8 (15 pts)** Metal conduits or hollow pipes are used in electrical wiring. In testing 1-inch pipes, these data are obtained on the outside diameter (in inches) of the pipe:

|       |       |       |       |
|-------|-------|-------|-------|
| 1.281 | 1.293 | 1.287 | 1.286 |
| 1.288 | 1.293 | 1.291 | 1.295 |
| 1.292 | 1.291 | 1.290 | 1.296 |
| 1.289 | 1.289 | 1.286 | 1.291 |
| 1.291 | 1.288 | 1.289 | 1.286 |

- (i) (5 pts) Find  $\bar{x}$ ,  $s^2$ , and  $s$  for this sample.
- (ii) (5 pts) Assume that sampling is from a normal distribution. Find a 95% confidence interval on the mean outside diameter of pipes of this type.
- (iii) (5 pts) The makers of this type of pipe claim that the mean outside diameter is 1.29 inches. Does the confidence interval lead you to suspect this reported figure? Explain.